Investra

A beginner-first systematic multi-asset investment app

FINS5545 FinTech Project. Part B

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Introduction

Investra is built first for a beginner, and everything the beginner sees is kept plain and free of jargon. For a more experienced investor the app also offers an advanced analytics view, reached from the Analytics tab, which gathers the methods and the evidence behind the funds in one place, so the product serves two readers at once without crowding either. The focus throughout is helping a first-time investor act with confidence.

→ See Screenshot 1 (Appendix).

Rather than asking a beginner to pick individual assets, Investra offers professionally built systematic funds. It pairs quantitative portfolio optimisation with investor education, so a user understands both the reason for each fund and the risk it carries. The app is the final stage of the Data Factory Floor, where clean data and models become an interactive product.

The funds are built from three universes, equity, cryptocurrency, and a combined multi asset universe, using systematic construction. Inside the app the investor can compare funds, examine past performance, test allocations, and read plain explanations before deciding.

Investra targets two barriers that stop new investors. The first is complexity, since terms like volatility, the Sharpe ratio, and optimisation put people off. The second is choice overload, since there are too many products to weigh up. The app answers both by guiding the investor through a clear journey, recommending funds that fit their profile, and explaining everything in plain language with simple visuals.

This report is written by following customer journey in the app. It begins on the Home page, which states what Investra offers and lays out the path ahead. From there the investor answers a short questionnaire on the Find Your Fund page, receives a recommended match, compares the nine funds, reads a fund fact sheet, simulates a future outcome in dollars, builds a blended allocation, reviews the market sentiment, and can look up any term in the Dictionary. Each section below follows one of these steps, explains the method and the formulas behind it, and points to the matching screenshot in the appendix.

1. Find Your Fund , Profiling the Investor

→ See Screenshot 2 (Appendix).

The journey starts with a short profile. The investor answers four simple multiple choice questions covering their goal, their time horizon, how they would react to a loss, and whether they want cryptocurrency. Asking about behaviour works better than asking directly how much risk they want.

The first three questions add up to a risk score from zero to six. Because each question captures a different angle, the score reflects the investor from several sides rather than a single guess.

Risk tolerance has several parts. A long horizon gives time to recover from a fall, the goal sets the balance between protecting and growing capital, and the reaction to a loss shows whether a person can stay invested in a downturn. Combining the three gives a fuller read than one direct question.

The profile is then matched to the funds. The app scores how close each fund's risk is to the investor's, ranks them, and presents the three most suitable, the best match and two runners up.

Risk score	Fund risk	Fit
0 to 6	0 to 6	0 to 100
risk score = goal + horizon + reaction	$\text{fund risk} = 6 \times \dfrac{\text{fund vol} - \text{lowest vol}}{\text{highest vol} - \text{lowest vol}}$	$\text{fit} = 100 \times \left(1 - \dfrac{\lvert \text{risk score} - \text{fund risk} \rvert}{6}\right)$

Cryptocurrency carries an extra guardrail. Because crypto is far more volatile than equities, the app only recommends a crypto fund when the investor has asked for crypto and has a profile that can bear high risk. This keeps a cautious or short horizon investor away from funds that can fall very sharply.

2. Your Match , Construction of the Investment Funds

Once the profile is set, the app recommends three funds, ranked as the best match and two runners up.

The product is not a single stock but an optimised portfolio built with quantitative methods.

Investment Universe	Portfolio Construction
Equity-only portfolios	Maximum Sharpe Ratio
Cryptocurrency-only portfolios	Minimum Variance
Combined multi-asset portfolios	Risk Parity

Together this gives a nine fund menu that spans a wide range of risk and return while keeping one consistent, systematic approach.

Maximum Sharpe Portfolio

The maximum Sharpe fund aims for the highest reward for risk. It weighs both the expected returns and how the assets move together, and picks the mix with the best return for each unit of risk.

Sharpe ratio	Max-Sharpe fund
reward for risk	choose weights w to maximise
$\dfrac{R_p - r}{\sigma_p}$	$\dfrac{w^T\mu - r}{\sqrt{(w^T\Sigma w)}}$

$R_p = w^T\mu$ is the blend return, $\sigma_p = \sqrt{(w^T\Sigma w)}$ is the blend risk.

Long only and fully invested: each weight ≥ 0 and the weights sum to 1. Investra uses a risk-free rate $r = 0$.

Because it favours the strongest performers, this fund often becomes concentrated, placing large weights on a few assets rather than spreading evenly.

As a result it tends to earn the highest long run returns but also the deepest falls. Combined Max-Sharpe, for example, had the highest return at 29.3 percent and the highest Sharpe at 1.14, but also the worst drop at minus 25.3 percent (Table 1).

Minimum Variance Portfolio

The minimum variance fund ignores expected returns and only tries to make the portfolio as steady as possible.

It spreads capital across assets that together give the smallest overall variance, which makes it well diversified. It earns less but moves far less, with smaller falls. Combined Min-Variance, for instance, returned 5.7 percent at 12.8 percent volatility, the lowest risk of the combined funds (Table 1).

This suits a cautious investor who values protecting capital over chasing growth.

Risk Parity Portfolio

Risk parity sits between the two. Instead of weighting by expected return, it sets the weights so every asset adds about the same amount of risk. That usually gives more diversification than maximum Sharpe and more return than minimum variance.

Risk contribution

each asset's share of the total risk

$$RC_i = \frac{w_i\,(\Sigma w)_i}{\sigma_p}$$

Risk-parity fund

weights set so every asset adds the same risk

$$RC_1 = RC_2 = \ldots = RC_n$$

$\sigma_p = \sqrt{(w^T \Sigma w)}$ is the blend risk, and the contributions sum to it: $RC_1 + \ldots + RC_n = \sigma_p$.

Steady assets get more weight, volatile assets get less. Long only, fully invested, weights sum to 1.

Risk parity is therefore a balanced choice for an investor who wants moderate risk and moderate growth.

Out-of-Sample Backtesting

This project uses a walk forward out of sample backtest. Rather than fitting the portfolios on the whole dataset at once, the weights at each date are estimated only from the information available up to that point, and they are then tested on the next, unseen period.

This approach prevents look-ahead bias, where future information accidentally influences investment decisions during model construction. By restricting each optimisation to information that would genuinely have been available to investors at that time, the reported performance more closely reflects achievable real-world investment outcomes. In this implementation the weights are estimated on a 252 day trailing window and rebalanced monthly on the first trading day, with a risk free rate of zero. Equity and combined funds are annualised using 252 trading days and crypto funds using 365, over an out of sample period from October 2020 to December 2023.

→ See Table 1 (Appendix).

→ See Figure 1 (Appendix).

3. Compare , Reading the Fund Fact Sheets

After the recommendation, the investor can compare the funds at two levels of detail. The Compare menu gives a straightforward view for a beginner, and the Fund fact sheet gives a fuller view for a more experienced investor. Offering both keeps the app flexible for new and intermediate investors alike.

Compare menu

In the Compare menu, Investra pairs the historical performance metrics with simple visualisations and plain explanations, so an investor can grasp the practical meaning of risk and return. The funds are grouped into three tiers. The first tier holds the safer and steadier funds with the lowest drawdowns. The second holds the balanced funds that trade return against drawdown. The third holds the higher risk funds with the highest returns. This lets a new investor make a basic comparison without being overwhelmed by technical detail.

→ See Screenshot 3 (Appendix).

Fund fact sheet menu

A more experienced investor can open the Fund fact sheet for a deeper comparison. The page summarises each fund using several widely used measures, the annualised return, the annualised volatility, the Sharpe ratio, and the maximum drawdown. Together these give a balanced picture of performance. The annualised return measures long term growth, the annualised volatility measures how much the value moves, the Sharpe ratio measures reward relative to risk, and the maximum drawdown measures the largest fall from a previous peak, which gives an intuitive sense of the losses an investor might face in a severe downturn.

The fact sheet also compares the fund with cash, using the risk free rate as the cash benchmark. The growth of the fund and the growth of cash over the three years are drawn on one graph so the investor can see the difference at a glance. The investor can also view the current holdings of the fund.

→ See Screenshot 4 (Appendix).

For beginner investors, maximum drawdown is often easier to interpret than volatility. While annualised volatility describes statistical variation in returns, maximum drawdown illustrates the largest decline that an investor would have experienced historically. This metric therefore helps users understand the emotional and financial impact of market downturns before investing.

Portfolio Weight Information

Every fund is rebuilt each month. The full history of these weights was produced during analysis and is included as Figure 3 for completeness. However, the application deliberately does not present a weights over time chart. A stacked chart of sixty holdings across three years is cluttered and difficult for a first time investor to read, and the history of past weights is not information required in order to choose a fund. What matters to the investor is what the fund holds now and how spread out those holdings are. For this reason the composition is presented as a simple table on the fund fact sheet under What it holds, which lists the target weights from the most recent rebalance, while a Technical detail panel reports the share held in the top three assets together with the Herfindahl index of the holdings.

→ See Figure 4 (Appendix).

→ See Figure 2 (Appendix).

→ See Figure 3 (Appendix).

4. Simulate , Projecting Investment Outcomes Honestly

→ See Screenshot 5 (Appendix).

Past performance shows how a strategy has behaved, but it cannot promise future returns. The Simulate page accepts this and instead shows a range of possible futures.

The investor picks a fund, a starting amount, a monthly contribution, and a horizon. A Monte Carlo simulation then explores many possible outcomes rather than one fixed forecast.

It models future value with a lognormal process built from each fund's historical annual return and volatility, generating many paths to estimate the spread of possible wealth over the horizon.

Unlike a projection based only on an average return, the Monte Carlo simulation takes uncertainty seriously. Each run is one possible future, drawn from the randomness of markets. The app summarises the many runs into three outcomes, an optimistic case, a median case, and a pessimistic case, so the investor can see the range of possible growth and decide with that range in mind.

In the chart these appear as the tenth percentile, the median, and the ninetieth percentile paths, which stand for the pessimistic, likely, and optimistic cases.

One simulated month	**The projection**
a lognormal step with a monthly contribution	percentiles taken across thousands of paths
$V(t) = V(t-1) \times \exp(m + sZ) + C$	worst = P10 · likely = P50 · best = P90

Z is a standard normal draw, m and s are the monthly drift and volatility from the fund annual figures, and C is the monthly contribution.
Values are shown after the management fee.

The Monte Carlo projection, one simulated month and the percentiles taken across thousands of paths.

Why use a Lognormal Model

The lognormal model is used because value cannot go below zero and prices compound rather than add, which makes it a standard choice for long run growth. It keeps the simulated values sensible while preserving the effect of compounding.

Making Volatility Tangible

The simulator also teaches risk. A high return fund shows much wider bands than a steady one, so the investor sees at once that a higher expected return comes with more uncertainty. This matters most for the crypto funds, where a tempting past return can hide a very wide range of possible outcomes.

5. Build Allocation , Diversification Made Visible

→ See Screenshot 6 (Appendix).

The Build allocation page lets the investor combine the nine funds into a portfolio of their own. After choosing a recommended fund, they can blend it with others, simulate the result, and read a fact sheet for the blend. The feature may look more advanced, yet it is built to stay simple and easy to use. It extends the journey beyond a single fund and introduces one of the most important ideas in finance, which is diversification.

Rather than encouraging investors to concentrate all capital within one investment strategy, the allocation builder demonstrates how combining funds with different characteristics can improve the overall risk-return profile of an investment portfolio. Users may select predefined allocation templates or manually adjust portfolio weights while immediately observing the resulting changes in expected return and portfolio risk.

As a starting point, the investor is presented with three basic options of portfolio. They are conservative, balanced, and growth. The three presets are ready made starting mixes for different comfort levels. Conservative leans on the steadiest funds, so its swings are smaller and its expected growth is lower. Balanced spreads across a wider set of funds and sits in the middle on both growth and risk. Growth leans towards the higher returning funds, including more of the crypto and maximum Sharpe funds, so it aims for more growth and accepts larger ups and downs. Each preset is only a starting point, and the investor can move the sliders to fine tune the mix and watch the expected return and risk update.

Calculating Portfolio Return

The expected return of the blended portfolio is calculated as the weighted average of the expected returns of the selected funds. Mathematically, this is expressed as

$$R_p = w^T \mu$$

where w represents the vector of portfolio weights and μ denotes the expected return of each investment fund.

This calculation is relatively intuitive. If an investor allocates equal capital to two funds with expected annual returns of 10% and 20%, the blended portfolio would have an expected return of approximately 15%.

Consequently, users can readily understand how changing portfolio weights influences the overall expected growth of their investments.

Calculating Portfolio Risk

Portfolio volatility cannot be calculated using a simple weighted average. The total portfolio risk depends not only on the volatility of each individual fund but also on how the funds move relative to one another over time.

The application therefore computes portfolio volatility using the covariance matrix of fund returns.

$$\sigma_p = \sqrt{(w^T \Sigma w)}$$

where Sigma denotes the covariance matrix derived from the historical relationships among the systematic investment funds. This formulation captures the interaction between assets and allows the application to quantify the benefits of diversification accurately.

Demonstrating Diversification and Simulation

The allocation builder compares naive risk with diversified risk directly.

Many beginners assume that holding several investments simply averages their risk. It does not. If two investments do not move perfectly together, combining them can push the blend's volatility below the average of its parts.

The Combined funds show this in practice. Equity and crypto are only loosely correlated, so a strong spell in one can offset a weak spell in the other. The app draws the chosen mix as a donut with updated estimates of return and volatility. The investor sees that diversification can lower risk while keeping most of the return. The figures are plain, the yearly growth, how bumpy it is, the reward for risk, and the worst drop. The blend can then be simulated in the same way as a single fund.

6. Sentiment , News as Context, Not a Trading Signal

→ See Screenshot 7 (Appendix).

The application applies the VADER tool to the news headlines. VADER gives each headline a score for how positive or negative it sounds. The company scores are averaged, equal weight, into a score for each sector. A sector day with no headlines carries the previous value forward, and a sector counts as neutral, at zero, before its first headline. The result is a simple measure of sector mood over time, shown as Figure 5. The original punctuation and capitalisation are kept, because they affect the score, so a headline that shouts in capitals can read differently from the same words in lower case.

Standardising Sentiment against its Own Normal

A sentiment tool such as VADER rates business news as mildly positive on average, so the baseline for a normal day sits above zero rather than at zero. Plotting the raw score therefore makes the news appear positive almost all of the time, even when nothing unusual has happened. To correct this bias, each sector score is converted into a standardised score, known as a z score. The z score subtracts the sector's own historical mean from the latest value and divides the result by the sector's historical standard deviation. A z score of zero means the sector sits exactly at its own normal level, a positive z score means the mood is above that normal, and a negative z score means it is below, with the distance measured in standard deviations. This standardisation adjusts the sentiment index so that it reflects how unusual the current mood is rather than how positive the raw number happens to look. The mood gauge and the sector labels read from these standardised values and fall into bands, where a size below one half of a standard deviation reads as about normal, a size between one half and one and a half reads as a little above or below normal, and a size beyond one and a half reads as unusually positive or negative. Because the baseline is removed, a sector can carry a positive raw score and still sit below its own normal. After this adjustment the gauge reads about normal rather than very positive, which supports the treatment of sentiment as context rather than as a buy or sell signal.

Preventing Look-Ahead Bias

A critical design decision is a minimum one day lag between published news and investment decisions. On any trading day t, portfolio construction uses only information available up to the previous day, day t minus one. This ensures that portfolio decisions do not incorporate information that an investor could not have observed at the time. Without this lag the backtest would suffer from look-ahead bias and report performance that could never be achieved under genuine market conditions. Maintaining this chronological consistency strengthens the credibility of the empirical results.

Sentiment Fusion

The module investigates whether portfolio optimisation can be improved by tilting allocations towards sectors that show more positive sentiment. A tuning parameter k controls the strength of the adjustment. When k = 0, portfolio construction ignores sentiment entirely and relies only on quantitative optimisation, and larger values of k increase the influence of sentiment on the allocation. Several values of k are evaluated to determine whether incorporating sentiment improves investment performance.

Experimental Results

The results show that increasing the sentiment tilt does not improve portfolio performance. The highest reward for risk is obtained when sentiment is ignored, at k = 0, and progressively larger tilts produce steadily lower Sharpe ratios, falling from 0.749 at k = 0 to 0.717 at the strongest tilt tested, k = 2 (Figure 6). This is one of the most informative outcomes of the project. Rather than justifying the sentiment module simply because it is technically appealing, the empirical evaluation shows that the additional information does not improve investment decisions within the systematic framework.

Interpretation through Market Efficiency

The result fits the semi-strong form of the Efficient Market Hypothesis. Public information, including the news, is priced in quickly, so by the time an investor reacts to a headline the move has usually already happened. Tilting towards sectors with good news therefore buys in after the rise. The honest conclusion is that sentiment is best used as context, not as a trading signal, and reporting a clear negative result is itself a useful contribution.

→ See Figure 6 (Appendix).

7. Dictionary , Investor Education

→ See Screenshot 8 (Appendix).

Investra puts education inside the workflow rather than assuming prior knowledge.

Unfamiliar terms carry tooltips that explain them in plain English, and a dedicated glossary covers ideas like volatility, the Sharpe ratio, drawdown, diversification, and optimisation.

Fund names also carry short popovers that explain the method behind each portfolio in everyday language, focused on what it means for the investor.

8. Design System and Implementation

Beyond the models, Investra was designed to feel professional and approachable for a first time investor, pairing a consistent look with a simple, reliable deployment. The identity is built around green, which reads as growth, stability, and trust, a natural fit for an investing app aimed at beginners. The layout is clean and consistent, and each page covers a single step of the journey so the investor moves through it logically. Clear charts, simple navigation, and responsive layouts help, and tooltips and short explanations keep the investor informed without needing prior expertise.

The aim throughout is to lower the mental barriers a beginner faces, so they can engage with each decision with confidence.

9. Advanced Analytics for the Experienced Investor

The Analytics tab is a dedicated home for the deeper work, kept off the beginner path so it never crowds a first time user. It brings together the efficient frontier with the nine funds plotted on it, the full metrics table, a correlation view of how the funds move together, the sentiment tilt sweep, the finVADER comparison, and the Fear and Greed method. This is where the depth of Investra is on show. The honest findings, the sentiment tilt that did not beat the baseline and the crypto maximum Sharpe fund that overfitted its history, are placed here as evidence rather than hidden. Showing these negative results is what gives the systematic funds their credibility, because it shows that the app trusts tested evidence over a reaction to the latest news.

Each element on the page answers a different question. The efficient frontier shows the best trade off between risk and return that the funds can reach, and where each of the nine funds sits against it. The correlation view shows which funds move together and which move apart, which is what makes a blend on the Build allocation page lower the overall risk. The full metrics table lists the annualised return, the volatility, the Sharpe ratio, and the worst drop for every fund side by side. The sentiment tools, the tilt sweep, the finVADER comparison, and the Fear and Greed method, gather the sentiment work and its result in one place. Keeping these on a separate page means the beginner journey stays simple, while a more experienced investor still has the full evidence to hand.

The Analytics page is shown in Screenshot 9 in the appendix.

Innovations at a Glance

Innovation	Description	Evidence in the app
Nine-fund framework	3 universes × 3 methods	Recommendation & Compare
Suitability recommender	Profile scoring, ranks funds, guards crypto	Find Your Fund + Compare
Monte Carlo simulator	Lognormal projection, probability bands	Simulate page
Allocation builder	Diversification via $w^T\mu$ and $\sqrt{(w^T\Sigma w)}$	Build Allocation
Sentiment module	Sector VADER index, empirical tilt test	Sentiment page
Investor education	Glossary, tooltips, plain terms	Dictionary + tooltips
Original design system	Coherent green Investra identity	Whole app
finVADER lexicon	VADER extended with finance and crypto word lists	Sentiment page and lexicon comparison
Fear and Greed index	finVADER scores scaled and standardised into mood bands	Sentiment gauge and fear_greed_index.csv
Cash benchmark	Kenneth French risk free rate as a comparison with a savings account	Fact sheet and vs_cash.csv
Management fee	0.75 percent a year, projections shown net of the fee	Simulate page
Efficient frontier	Long only mean variance frontier with the funds plotted	Analytics page and efficient_frontier.csv
Analytics view	A dedicated advanced home, progressive disclosure	Analytics page

10. Critical Reflection and Recommendations

Project Reflection

Investra shows how quantitative finance, data analytics, and user centred design can come together in one platform for beginners. Some parts worked well, and others exposed limits worth improving.

The strongest part is the systematic construction. Three methods across three universes let an investor compare risk and return without picking stocks, and the walk forward out of sample backtest keeps the results honest by never using future data.

The diversification framework is another success. The Combined funds show clearly that mixing loosely correlated assets can lower volatility while keeping returns competitive, turning an abstract idea into something the investor can see.

The Monte Carlo simulator adds to this by showing a range of outcomes rather than one number, which communicates uncertainty honestly and cools unrealistic hopes.

However, not every experimental component achieved the anticipated outcome.

The sentiment tilt did not improve performance. Adding sector news sentiment to the optimisation lowered the reward for risk. Disappointing at first, this is a useful result, because it shows the value of testing an idea rather than assuming that more information always helps.

The crypto maximum Sharpe fund also weakened out of sample. Heavy concentration in a few coins led to extreme falls, showing how unstable return estimates cause overfitting in very volatile markets.

These outcomes reinforce the broader lesson that sophisticated optimisation techniques remain dependent upon the quality and stability of their underlying data.

The recommendations below also draw on a comparison of Investra with award winning platforms such as moomoo, Tiger Brokers, CMC Invest, and eToro, and they target the gaps that comparison revealed.

Recommendation 1. Weight the News by Source Credibility

At present every headline is scored in the same way and counts the same amount, yet anyone can publish news and not all of it is factual or reliable. Award winning platforms such as CMC Invest pair the raw feed with vetted material, including analyst ratings and established newswires. A stronger version of Investra would add a research and values layer that gives more weight to credible and verified sources and less to unverified or low quality items, and that leans on trusted analysis rather than the whole undifferentiated stream. Filtering for credibility before the sentiment is measured would produce a cleaner signal and a more trustworthy reading of the market mood. It also speaks to one likely reason the sentiment tilt did not add value, that the input mixed reliable analysis with noise.

Recommendation 2. Connect Live Market Data with Watchlists and Alerts

Investra currently runs on a fixed backtest from 2020 to 2023, so its view of the market is always out of date. The leading apps from moomoo and Tiger Brokers (references) run on live data with watchlists and price alerts. Connecting a live data feed would let Investra refresh its metrics, its sentiment, and its fund performance as the market moves, and it would let the investor keep a watchlist and receive an alert when a fund's drawdown, volatility, or news mood crosses a level they have set. This would require a paid market data interface and would add a running cost, yet the payoff is a more current and more credible analysis and a clear reason for the investor to return.

Recommendation 3. Add a Practice Account that Tests the Funds on New Data

The reference broker give beginners a practice account where they invest simulated money and follow the result over time. Investra only shows a hypothetical projection, and its numbers come from a backtest that has already happened. A practice account would let the investor place a virtual allocation across the nine funds and then watch it play out on live, previously unseen data, which is a real forward test rather than a replay of the past. It would teach the habit of investing without any real risk, and it would also give honest out of sample evidence on whether the funds hold up on data they were never fitted to.

Conclusion

Investra shows how portfolio optimisation, education, and clear visualisation can combine in a beginner focused app. By replacing stock picking with systematic funds, it simplifies the decision while teaching the basics of portfolio management.

Overall it meets its goal of lowering complexity for new investors and supporting informed, evidence based decisions, and it shows how financial technology can improve access, understanding, and confidence for first time investors.

Appendix. Exhibits

Table 1. Out of sample performance of the nine funds over 2020 to 2023, annualised, before fees, with a risk free rate of zero.

Fund	Ann. return	Ann. vol	Sharpe	Max drawdown	$1 became
Combined Max-Sharpe	29.3%	25.7%	1.14	−25.3%	$2.16
Combined Risk-Parity	14.0%	16.0%	0.87	−19.8%	$1.48
Combined Min-Variance	5.7%	12.8%	0.45	−15.2%	$1.18
Equity Max-Sharpe	13.8%	18.4%	0.75	−23.3%	$1.47
Equity Risk-Parity	9.9%	14.6%	0.68	−18.5%	$1.33
Equity Min-Variance	5.6%	12.8%	0.44	−15.1%	$1.18
Crypto Min-Variance	82.7%	65.1%	1.27	−71.1%	$7.10
Crypto Risk-Parity	58.1%	78.7%	0.74	−79.6%	$4.43
Crypto Max-Sharpe	11.8%	75.5%	0.16	−81.6%	$1.44

Figure 1. Growth of $1 by fund family (log scale). Crypto Min-Variance compounds $1 to $7.10 and Combined Max-Sharpe to $2.16, versus $1.18 for the minimum-variance funds.

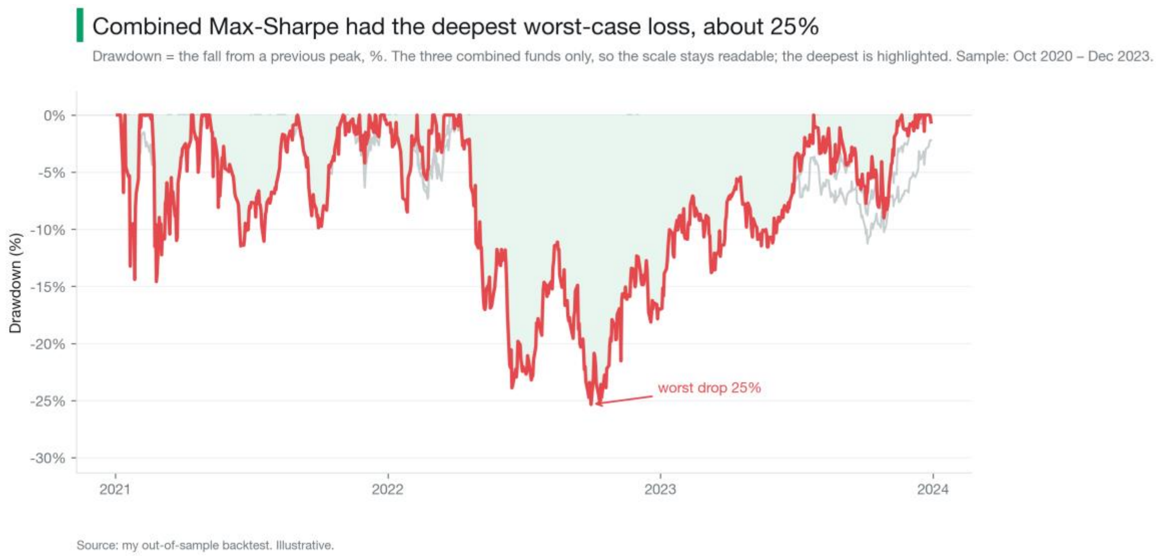


Figure 2. Drawdowns of the Combined funds. Combined Max-Sharpe suffers the deepest fall at −25.3%.

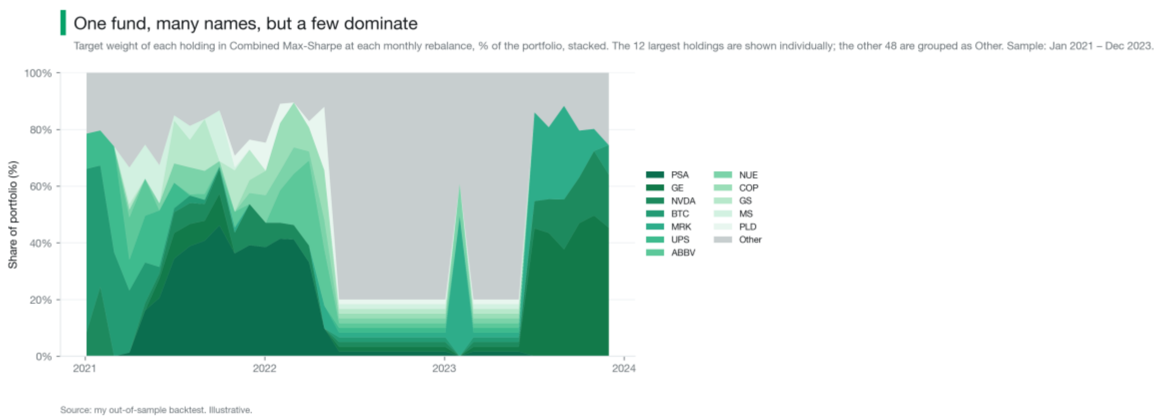


Figure 3. Portfolio weights over time for one fund, showing reallocation at each monthly rebalance.

Figure 4. Sharpe ratio across the nine funds. Crypto Min-Variance (1.27) and Combined Max-Sharpe (1.14) lead, and Crypto Max-Sharpe is weakest at 0.16.

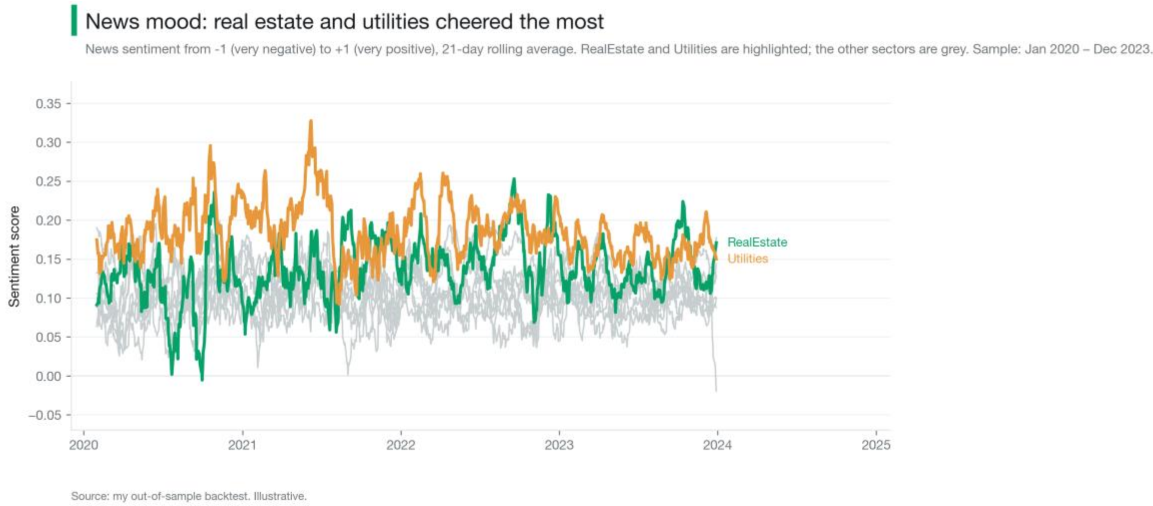
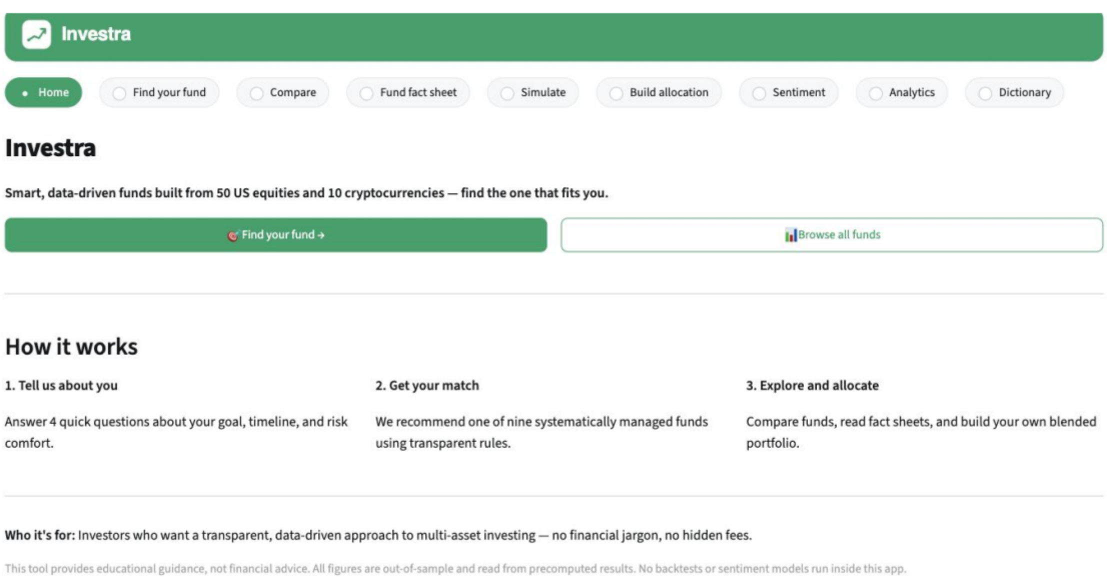
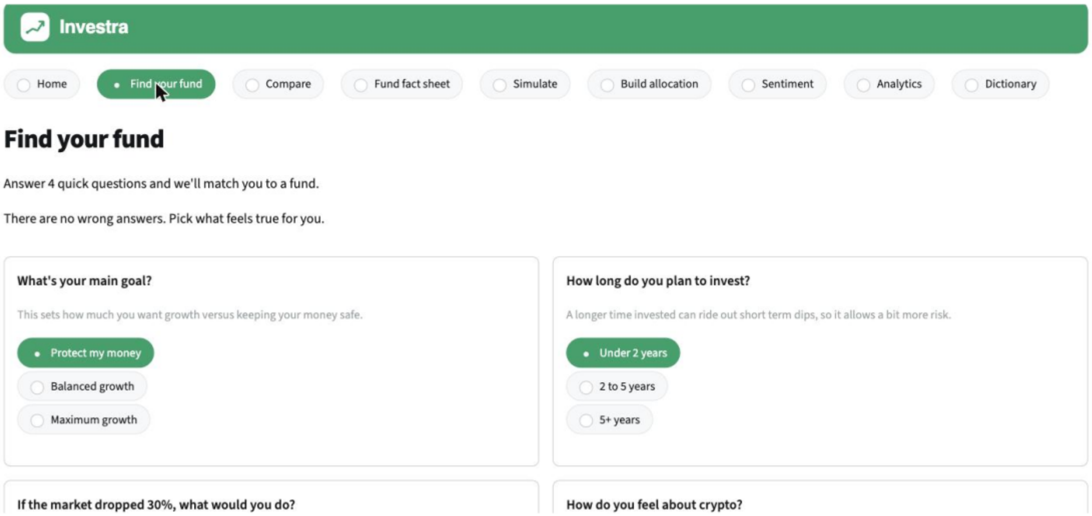


Figure 5. Sector sentiment index over time (VADER compound score, lagged ≥ 1 trading day).

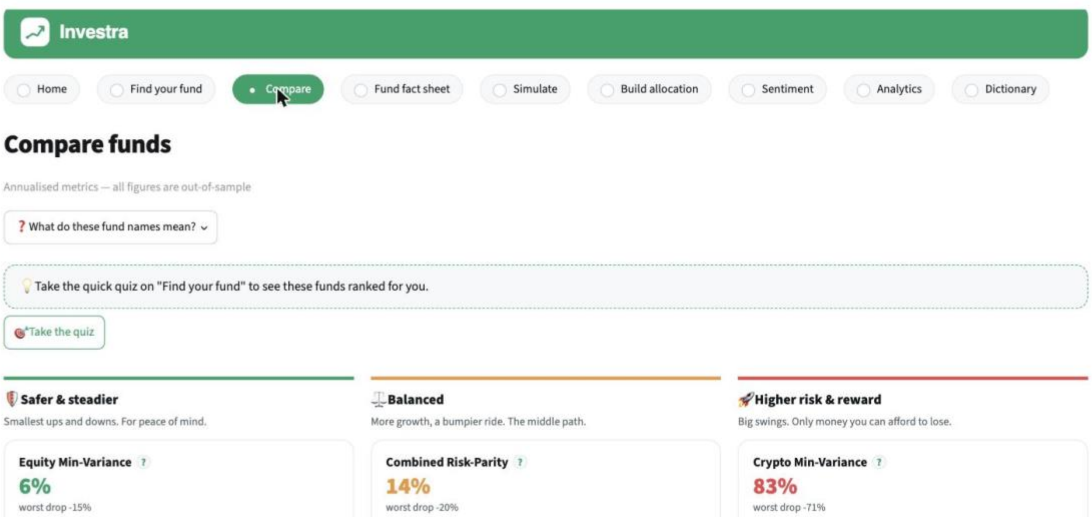
Figure 6. Fusion k-sweep. Reward-for-risk is highest at k = 0 (0.749) and declines monotonically to 0.717 at k = 2.



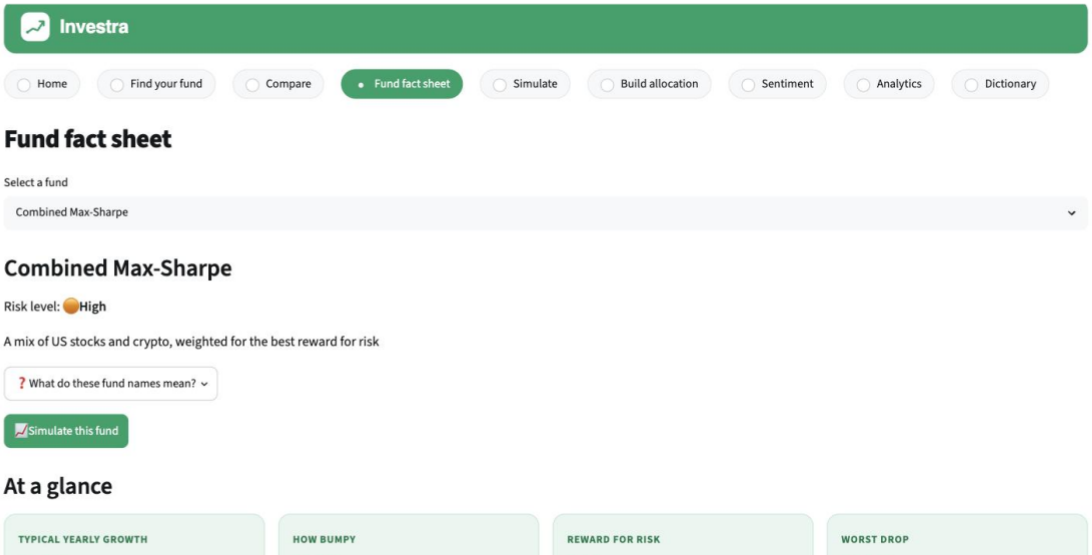
Screenshot 1. Investra home page, a guided entry point stating the value proposition and three step journey.



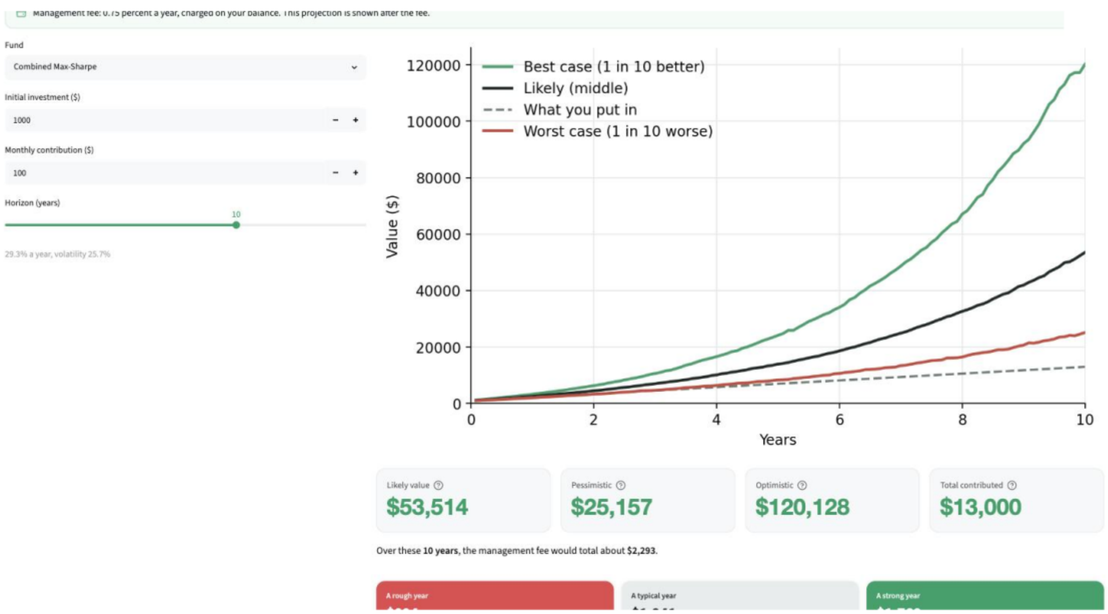
Screenshot 2. Find Your Fund, four plain language questions that build the investor's risk profile.



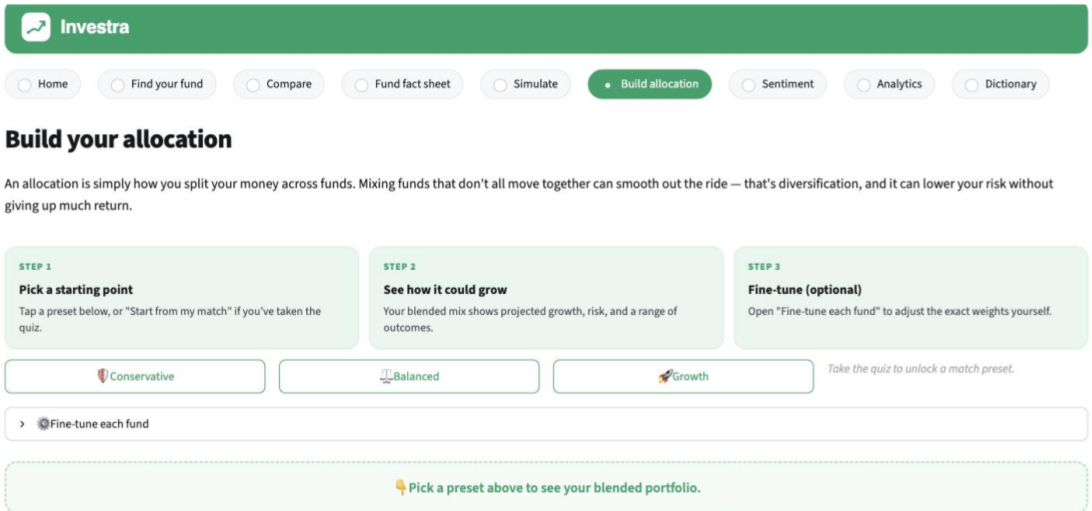
Screenshot 3. Compare groups the nine funds into three comfort tiers.



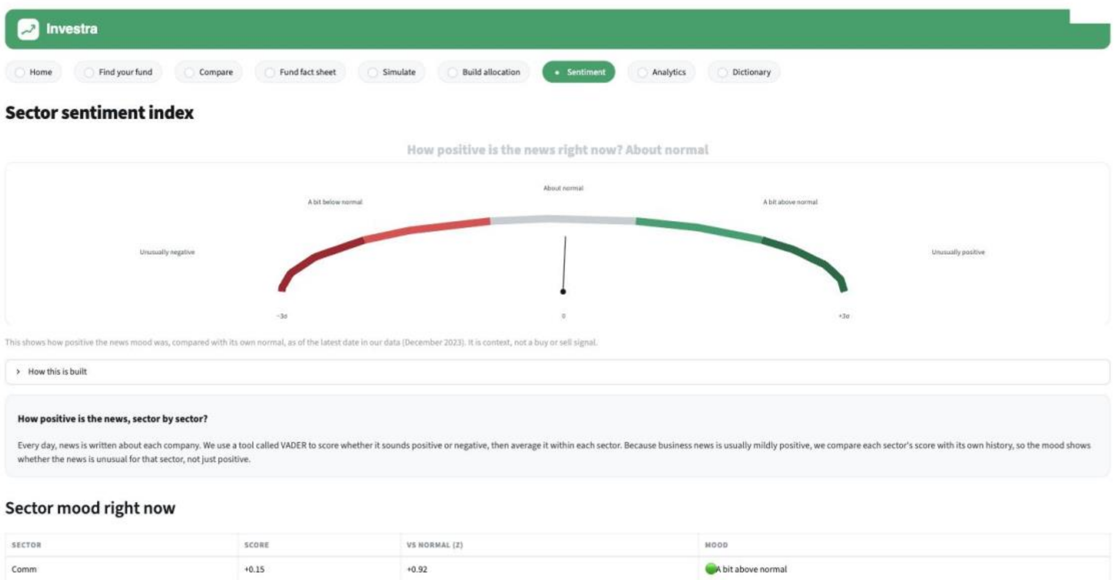
Screenshot 4. A fund fact sheet (Combined Max-Sharpe), metrics, growth, drawdown, and holdings.



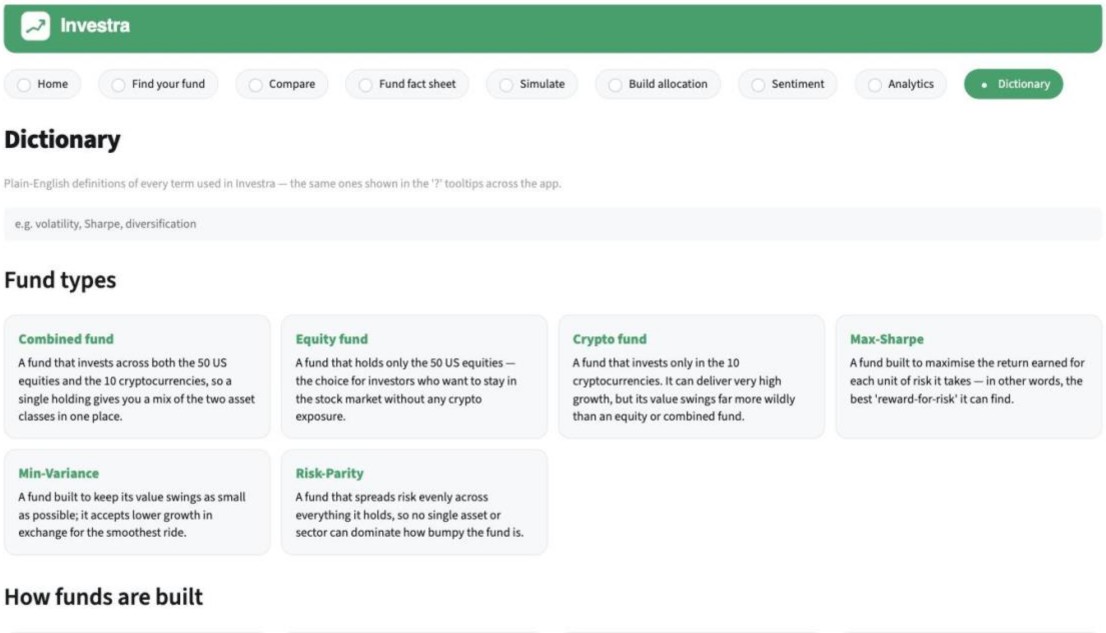
Screenshot 5. The Simulate page, the four-line projection showing the worst, likely and best cases against the amount contributed, with the projection values and the fee.



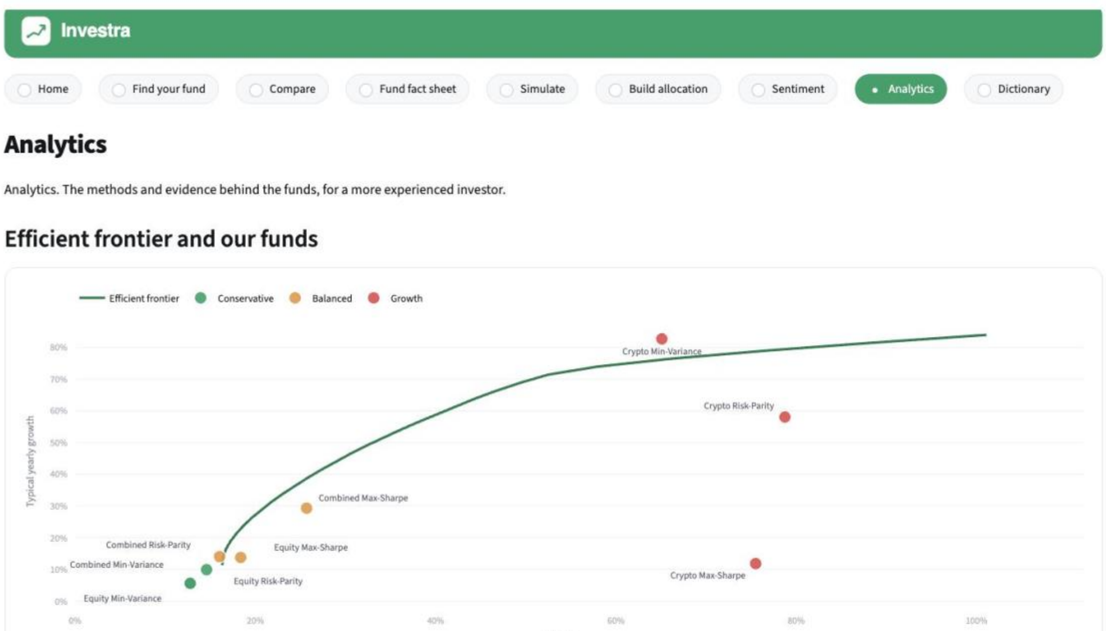
Screenshot 6. Build Allocation, plain language guidance and one-click presets.



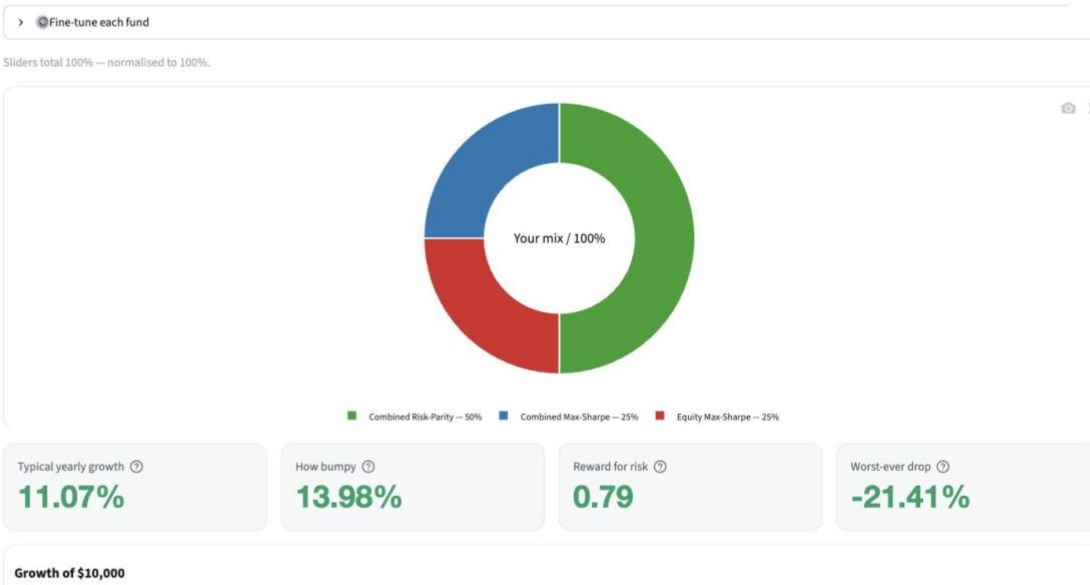
Screenshot 7. The Sentiment page, the mood gauge standardised against the news own normal, which now reads about normal rather than very positive.



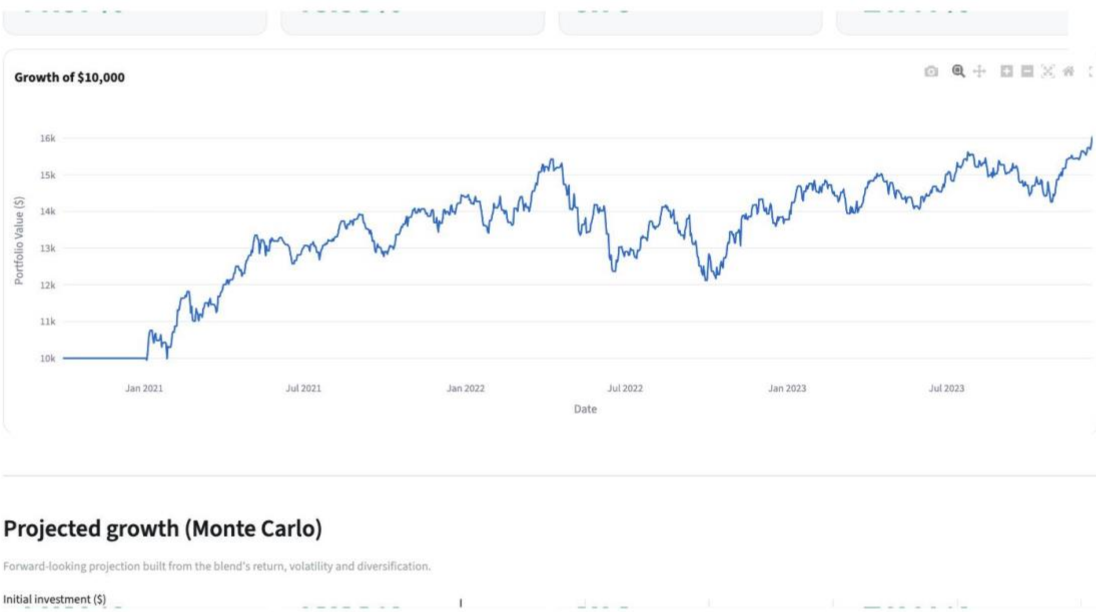
Screenshot 8. Dictionary, plain English definitions grouped by theme.



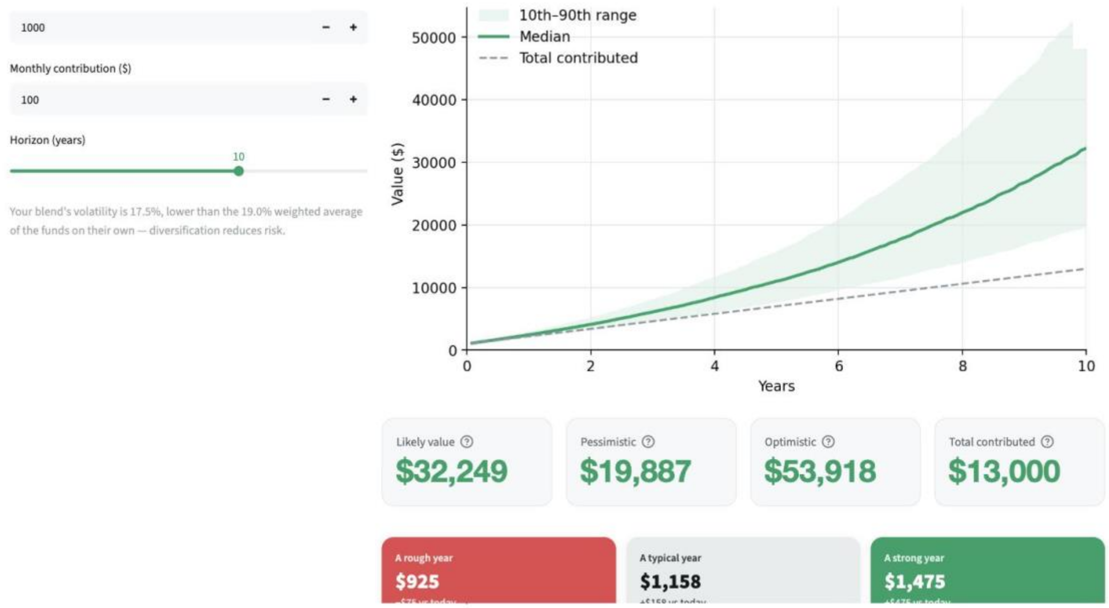
Screenshot 9. The Analytics page, showing the efficient frontier, the metrics table, the correlation view, and the sentiment deep dive.



Screenshot 10. Build Allocation after the Balanced preset, a blended donut across three funds with plain-language metrics for growth, bumpiness, reward for risk and worst drop.



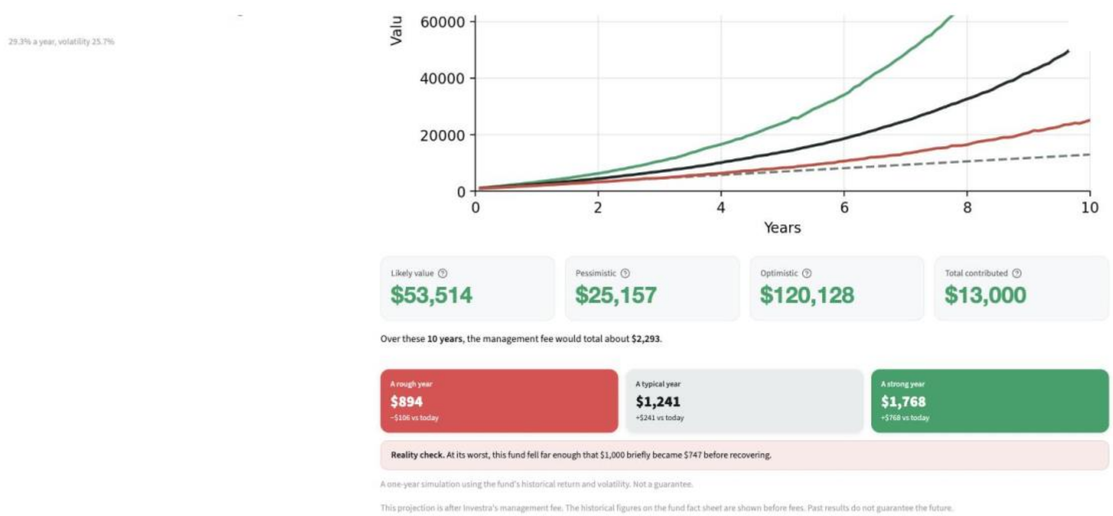
Screenshot 11. Build Allocation, growth of $10,000 for the blended mix across the out-of-sample period.



Screenshot 12. Build Allocation, the Monte Carlo projection with the diversification note that the blend's volatility of 17.5% sits below the 19.0% weighted average of its funds on their own.

Screenshot 13. Fund fact sheet, how your money grew against cash, where $1 became $1.18 in the fund versus $1.06 in cash, a lead of about 3.6 percent a year.

Screenshot 14. Fund fact sheet, the risk view, a drawdown chart showing $1 dipped to $0.85 at the worst fall of about 15 percent before recovering.



Screenshot 15. Simulate, the dollar scenarios for a rough, typical and strong year, with the management fee disclosed as about $670 over ten years.

-3σ 0 +3σ

This shows how positive the news mood was, compared with its own normal, as of the latest date in our data (December 2023). It is context, not a buy or sell signal.

› How this is built

How positive is the news, sector by sector?

Every day, news is written about each company. We use a tool called VADER to score whether it sounds positive or negative, then average it within each sector. Because business news is usually mildly positive, we compare each sector's score with its own history, so the mood shows whether the news is unusual for that sector, not just positive.

Sector mood right now

SECTOR	SCORE	VS NORMAL (Z)	MOOD
Comm	+0.15	+0.92	A bit above normal
Energy	+0.13	+0.70	A bit above normal
Industrials	+0.13	+0.60	A bit above normal
Healthcare	+0.12	+0.46	About normal
Financials	+0.10	+0.35	About normal
Utilities	+0.10	+0.08	About normal
RealEstate	+0.08	-0.11	About normal
Tech	+0.06	-0.16	About normal
Consumer	-0.00	-1.02	A bit below normal
Materials	-0.16	-1.63	Unusually negative

As of 29 Dec 2023. Equity sectors only. Crypto has no news signal. Score is the average finVADER headline score; vs normal (z) shows how far from that sector's own average it is, in standard deviations.

Does reacting to the news improve returns? Not in our tests.

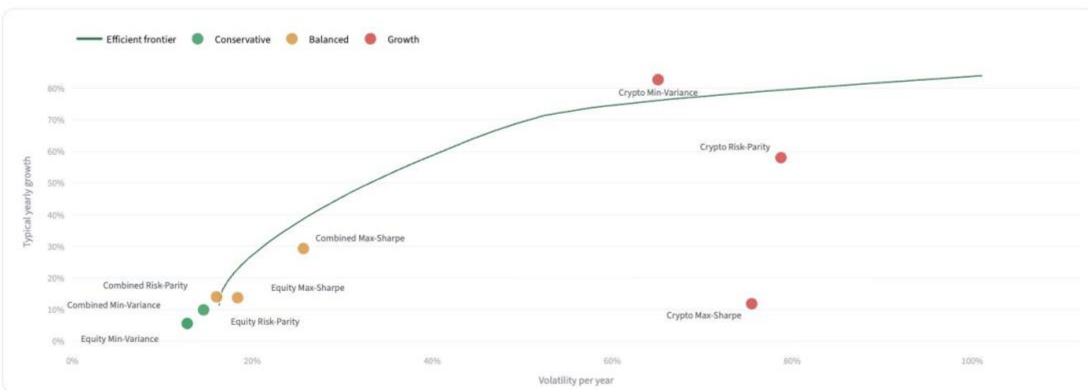
HOW MUCH THE FUND FOLLOWS THE NEWS	TYPICAL YEARLY GROWTH	REWARD FOR RISK (SHARPE)	STATUS

Screenshot 16. Sentiment, the sector mood table with a vs normal column, where a sector positive in raw terms can still sit below its own normal.

Analytics

Analytics. The methods and evidence behind the funds, for a more experienced investor.

Efficient frontier and our funds



The frontier is the best in sample tradeoff. Our funds are out of sample, so they generally sit inside it. Small deviations reflect the difference between the estimation period and the live period.

Full metrics table

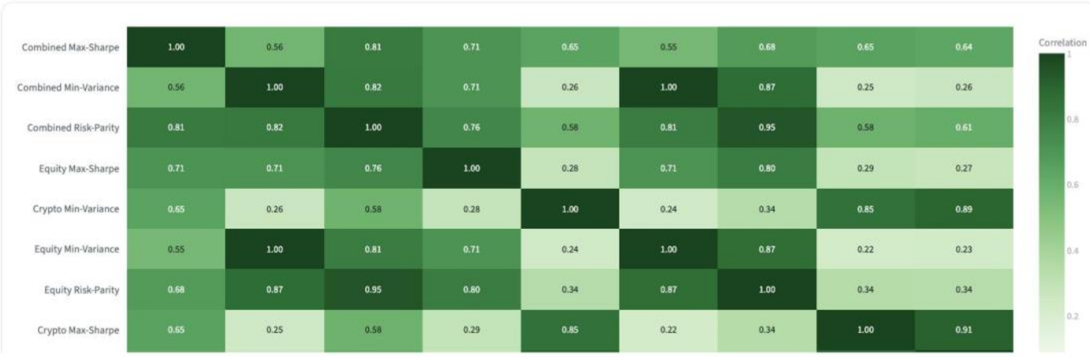
Screenshot 17. Analytics, the efficient frontier with the nine funds plotted by volatility and return and coloured by comfort tier.

Crypto Min-Variance	82.73%	65.09%	1.27	-71.09%	$7.10
Crypto Risk-Parity	58.07%	78.74%	0.74	-79.56%	$4.43

Out of sample backtest over 2020 to 2023, annualised and shown before fees.

The funds form a clear risk and return ladder. The minimum variance funds are the steadiest with the lowest returns, while the crypto funds carry the highest returns and the deepest falls.

How the funds move together



Screenshot 18. Analytics, the correlation heatmap showing how the nine funds move together.

Appendix. Methods and formulas

This appendix sets out how each figure in the app is produced, so the analytical process is fully transparent.

Annualisation and the Sharpe ratio

Daily returns are compounded into annual returns, and volatility is the standard deviation of daily returns times the square root of the number of trading days in a year, 252 for equity and combined funds and 365 for crypto. The reward for risk is the Sharpe ratio, the annual return minus the risk free rate, divided by the annual volatility. Investra states a risk free rate of zero, so here the Sharpe ratio is simply the annual return divided by the annual volatility.

Portfolio return and risk

A blended portfolio expected return is the weighted average of the fund returns, $Rp = w^T\mu$, where w holds the weights and μ holds the expected returns. A blended portfolio risk is not a weighted average. It is $\sigma p = \sqrt{(w^T\Sigma w)}$, where Σ is the covariance matrix, which captures how the funds move together and is why a diversified blend can carry less risk than the average of its parts.

The three portfolio construction methods

Each fund uses one of three methods. The maximum Sharpe method chooses the weights that give the highest reward for risk, the largest value of the annual return minus the risk free rate divided by the annual volatility. The minimum variance method ignores return and chooses the weights that give the smallest portfolio variance $w^T\Sigma w$, so it is the steadiest. The risk parity method sets the weights so that every asset contributes the same amount of risk, where the risk contribution of asset i is w_i times the i-th element of Σw divided by the portfolio volatility, which gives more weight to steady assets and less to volatile ones.

Correlation

The correlation between two funds is the Pearson correlation of their daily returns, the covariance of the two return series divided by the product of their standard deviations. It runs from minus one to plus one, and a value near zero means the two funds move almost independently, which is the source of diversification.

Turning the questionnaire into a match

The four questions become a match in two steps. First, three of the questions score the investor from zero to six. The main goal scores zero for protect my money, one for balanced growth, and two for maximum growth. The horizon scores zero for under two years, one for two to five years, and two for five years or more. The reaction to a thirty percent fall scores zero for sell, one for hold, and two for buy more. These add to a risk score from zero to six, and the fourth question records whether the investor wants crypto.

Second, each fund is scored for fit. Every fund volatility is placed on the same zero to six scale, where the fund risk equals six times the fund volatility above the lowest fund volatility, divided by the range of fund volatilities. The fit score is one hundred times one minus the distance between the investor risk score and the fund risk, divided by six, so a fund whose risk matches the investor scores near one hundred. Penalties then apply. A crypto fund loses fifty points if the investor chose stocks only, a crypto fund loses thirty points on a short horizon, and a risky fund loses twenty points for a very cautious investor. The highest scoring fund is the recommended match, and the same scores rank every fund from most to least suitable.

The sentiment tilt and the Fear and Greed index

The sentiment tilt multiplies each fund weight by one plus k times its sector sentiment and then renormalises, where k sets how hard the fund leans on the news and a k of zero ignores it. For the Fear and Greed index, each finVADER score is scaled from minus one to plus one onto a range of zero to one hundred, averaged per day, smoothed, then standardised against its own history so the bands measure greed or fear relative to normal.

Standardising sentiment with a z-score

Raw sentiment carries a positive baseline, so each sector score is standardised against its own history. The standardised value is the score minus the sector mean divided by the sector standard deviation, which is a z-score. A z-score of zero means the sector is at its own normal, and the mood bands are set on the z-score, where a size below one half of a standard deviation reads as about normal, between one half and one and a half reads as a little above or below normal, and beyond one and a half reads as unusually positive or negative.

The efficient frontier and excess over cash

The efficient frontier is the set of long only portfolios with the lowest volatility at each level of return, found by minimising $w^T\Sigma w$ at each target return. The excess over cash is a fund annual return minus the annual return of the risk free rate over the same period.

The Monte Carlo projection

The simulator grows wealth one month at a time with a lognormal random walk. Each month the value is multiplied by the exponential of a normal draw with monthly drift and monthly volatility taken from the fund annual figures, and the monthly contribution is added on top. Thousands of paths are generated, and the tenth, fiftieth and ninetieth percentiles across paths at each month give the worst, likely and best lines shown to the investor.

The management fee

Projected values on the simulator are shown after a management fee of 0.75 percent a year. The historical figures on the fund fact sheet are shown before fees, and this difference is stated on the page so the cost of the product is visible.